Simple Drive parabolic antenna
used in RadioSkyMap

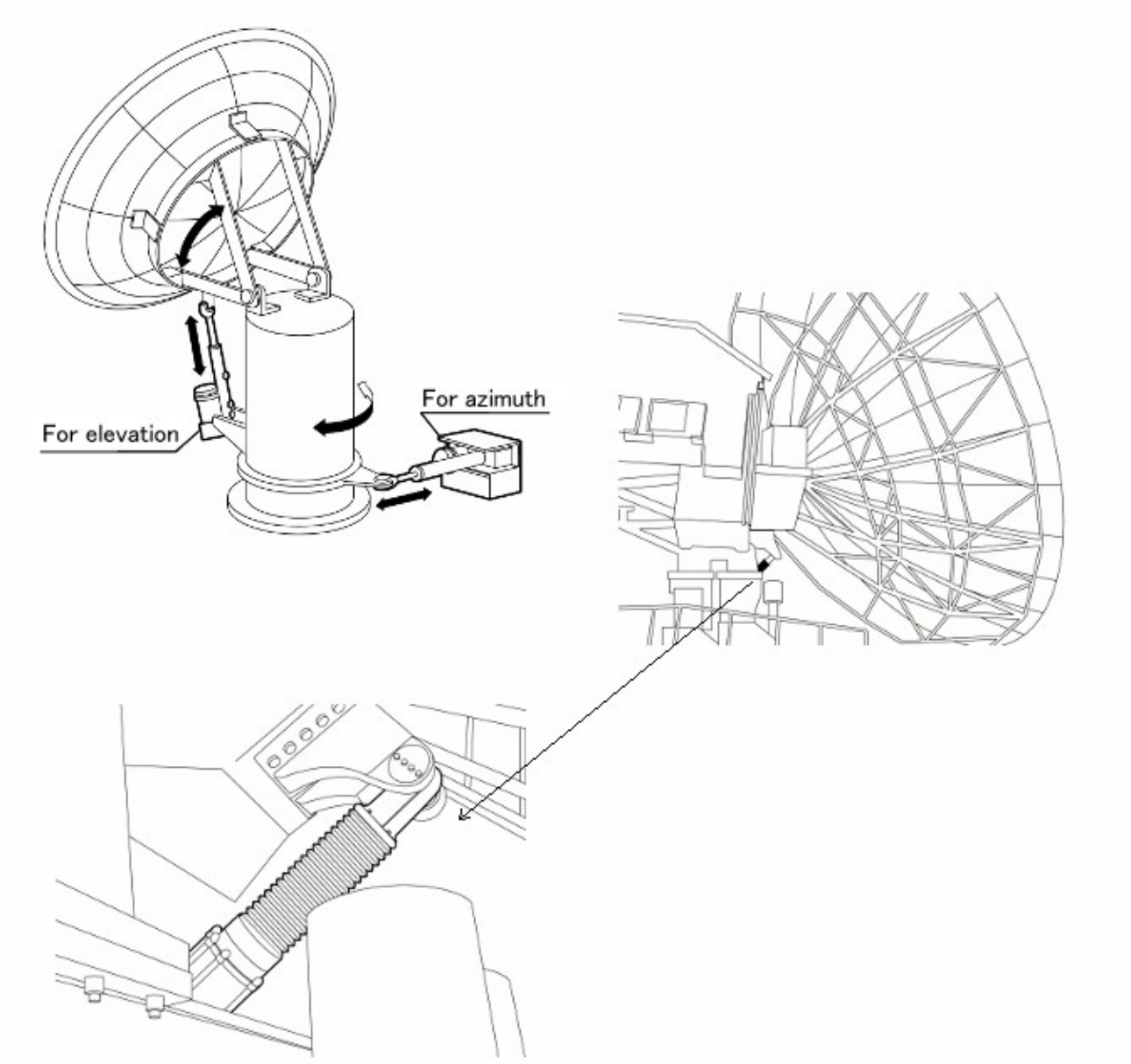
RadioSkyMap is able to control your telescope mount with the ASCOM universal telescope interface.
But for larger antennas or for those who do not have a control system for their antenna, it is possible to make a simple and low-cost system to control the pointing of your antenna at least in height. This system described below is implemented in RadioSkyMap and allows automatic pointing and observation in "Meridian Transit" mode or to make "drift scan" of the radio sky that can be used by the ezRA software for example.

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1. Antenna pointing by linear actuator

You can drive simply your parabolic antenna with linear actuator.
Below an excellent drawing of the principle from Japanese web site :

https://tt-net.tsubakimoto.co.jp/tecs/aplc/sad/aplc_sad_10.asp



2. Types of Linear actuator

Many linear actuator size exist now.
On my 3 meters antenna I have old and big linear actuator from original manufacturer (Strong USA) like below image :



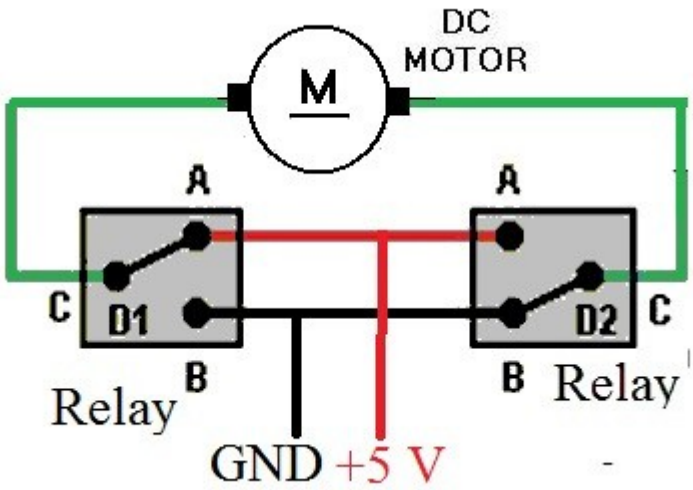
Is large, heavy and expensive…
But other solution exist now.
Many Chinese low price product exists, you can find theses linear actuator on Amazon or Aliexpress, with various power in N/m, various stroke, speed, power supply etc.

sample below :



3. How to drive theses linear actuator ?

The easiest solution.
DC power supply with dual switch, using schematic below :



Other solutions with dedicated electronic components exist. But in my opinion, this one is the simplest, the most economical and the most reliable in the long term.

4. USB relay used and implemented in RadioSkyMap

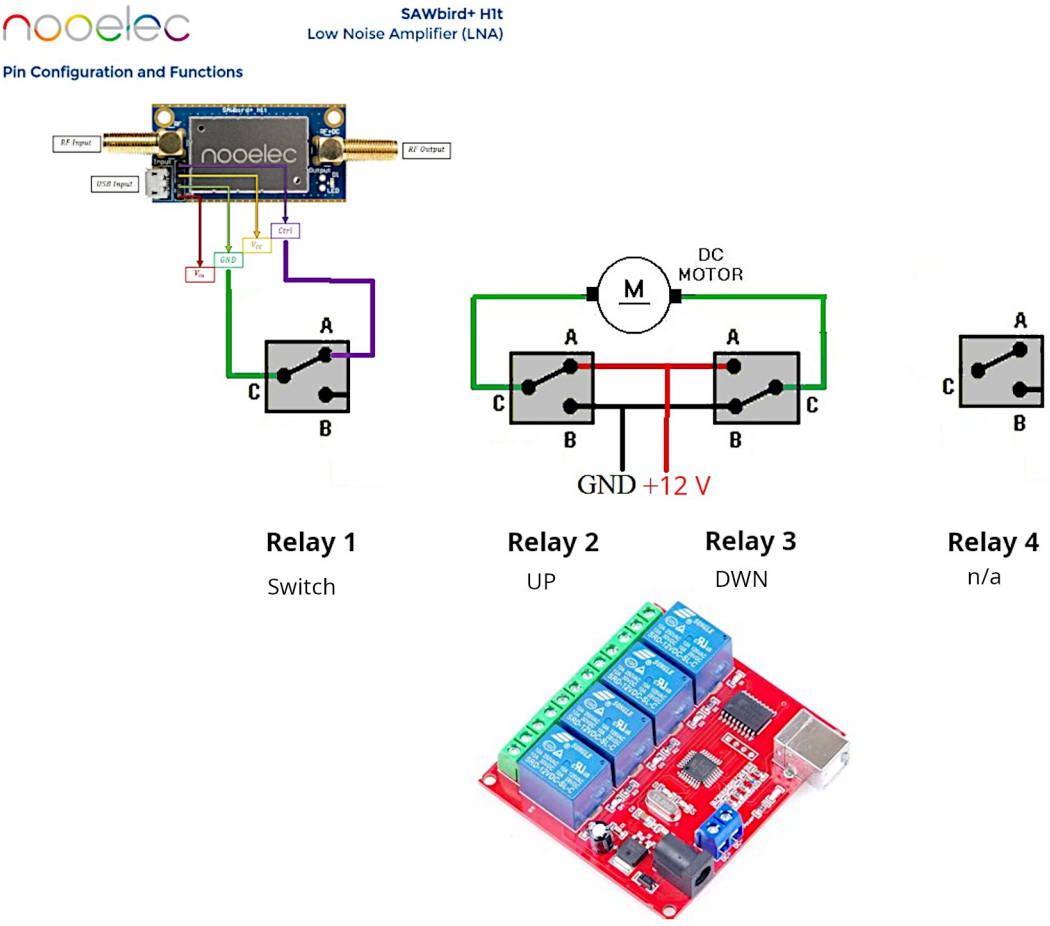
RadioSkyMap drive directly 2 to 8 relay through USB relay interface.
This interface is very common, inexpensive and can be easily found on Amazon or Aliexpress.



Simply connect it via a USB cable to your computer and RadioSkyMap will drive the actuator's motor.

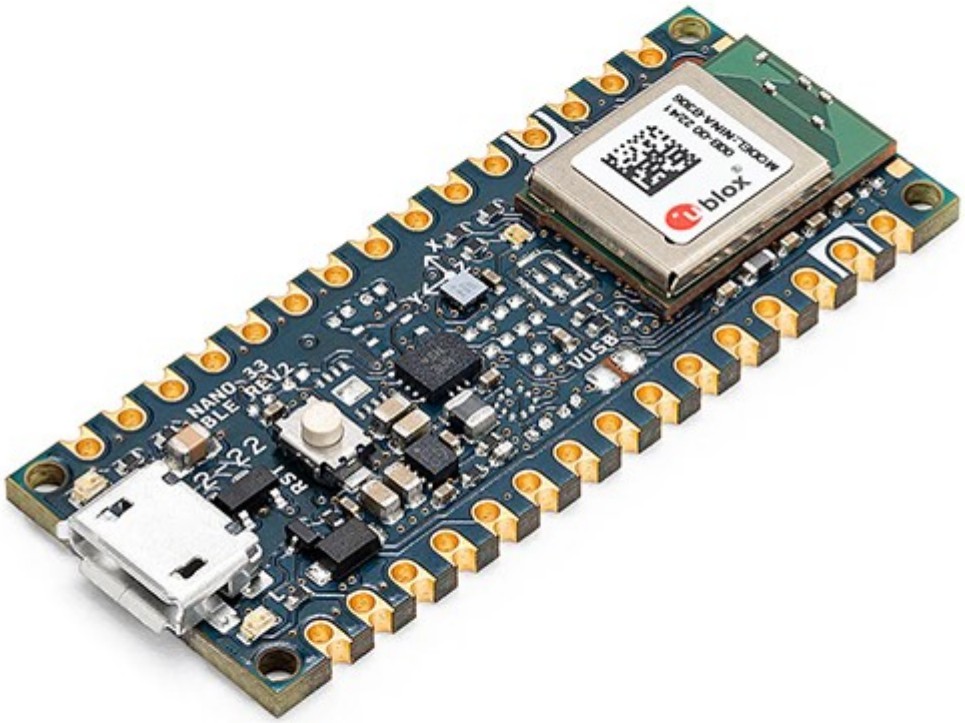
5. Relay wiring used in RadioSkyMap

Warning : respect relay number !



6. Antenna position control (Only in height) for meridian transit observations.

In order to simplify as much as possible the electronic realization of the antenna position control (at least in height), we will use an Arduino that directly integrates IMU-type position sensors. It is the Arduino nano 33 Ble.



So no soldering or complicated wiring, you just have to position the Arduino module correctly on the antenna and connect it by a USB cable to the computer that runs RadioSkyMap (as for the USB relay board) .
The electronic board is flat when the antenna is horizontal and normally when the antenna rises there is a change in the value of the IMU which measures the x position as the antenna rises. The code (sketch for Arduino IDE) below will transmit the x-position angle in degrees to RadioSkyMap on the USB port, which will be able to control the actuator motor to go to a position assigned in the software.

7. Arduino nano 33 Ble sketch to measure height angle (0.1° precision)

```
/*
  Arduino BMI270_BMM150 - Accelerometer Application

  This example reads the acceleration values as relative direction and degrees,
  from the BMI270 sensor and prints them to the Serial Monitor.
  The circuit:
  - Arduino Nano 33 BLE Sense Rev2
  Created by Pedro Lima
  This example code is in the public domain.
*/

#include "Arduino_BMI270_BMM150.h"

#define MINIMUM_TILT 0    // Threshold for tilt detection in degrees
#define WAIT_TIME 150     // How often to run the code (in milliseconds)

float x, y, z;
float angleX = 0;
int angleY = 0;
unsigned long previousMillis = 10;

void setup() {

  Serial.begin(9600);
  while (!Serial);

  if (!IMU.begin()) {
    Serial.println("Failed to initialize IMU!");
    while (1);
  }

  Serial.print("Accelerometer sample rate = ");
  Serial.print(IMU.accelerationSampleRate());
  Serial.println("Hz");
}

void loop() {
  if (IMU.accelerationAvailable() && millis() - previousMillis >= WAIT_TIME) {
    previousMillis = millis();
    IMU.readAcceleration(x, y, z);

    // Calculate tilt angles in degrees
    angleX = atan2(x, sqrt(y * y + z * z)) * 180 / PI;
    angleY = atan2(y, sqrt(x * x + z * z)) * 180 / PI;
    Serial.println(angleX);


  }
}
```

That all for drive your antenna in height.

For Azimut drive the encoder is necessary.
It is not implemented in RadioSkyMap at this time, because I whant to find inexpensive solution an a simple solution to implement by someone who is not necessarily an electronics engineer.

Pierre Terrier - 2026